

Portfolio Standard Deviation

Two asset portfolio:

$$\sigma_p = \sqrt{w_1^2 \sigma_1^2 + w_2^2 \sigma_2^2 + 2 w_1 w_2 \text{Cov}_{1,2}}$$

$$\text{COR}(X, Y) = \frac{\text{COV}(X, Y)}{\sigma_X \cdot \sigma_Y}$$

=correl(array1, array2)

Value between -1 to +1

Measure degree of linear relationship

Minimum Variance Portfolio

Weights		
Stock A	Stock B	Portfolio St. Dev.
100%	0%	9.43%
90%	10%	7.44%
80%	20%	5.46%
70%	30%	3.51%
60%	40%	1.74%
50%	50%	1.35%
40%	60%	2.95%
30%	70%	4.87%
20%	80%	6.84%
10%	90%	8.83%
0%	100%	10.83%



Some weighted combination of stocks A & B will provide a portfolio with minimized standard deviation

Not optimal

solver:

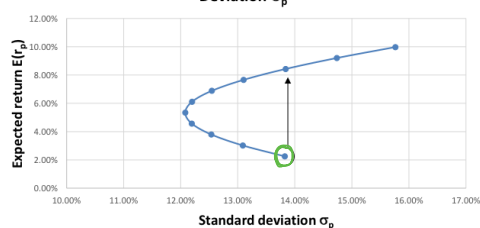
set "portfolio std. dev." to "min"
by changing "stock A weight"

B	C	D	E	F	G	H	I	J
0%	100%	10.83%						

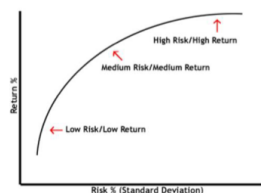
for Solver Standard Deviation Minimization		
Stock A	Stock B	Port. Std. Dev.
53.50%	46.50%	1.10%

Set Objective:		\$D\$3
To:	☐ Max ☒ Min ☐ Value Of:	0
By Changing Variable Cells:		\$B\$2
Subject to the Constraints:		
\$B\$2 <= 1		
\$B\$2 >= 0		

Portfolio Returns: Expected Return $E(r_p)$ and Standard Deviation σ_p



Efficient Frontier



Risk Free Asset

Consider a portfolio of two risky assets (stocks A & B) and a risk free asset -

a savings account, government bond, etc.

Let's assume $r_f = 2\%$

Effect of Correlation

With a correlation of +1, it is impossible to diversify away any risk

With a correlation between -1 and +1, some risk can be eliminated through diversification

With a correlation of -1, a risk free portfolio can be created using a combination of the two assets

Sharpe Ratio (return vs. risk)

$$\frac{(R_p - R_f)}{\sigma_p}$$

R_p = return on portfolio
 R_f = the risk-free rate of return
 σ = portfolio standard deviation

Portfolio A generates a return of 15%, $\sigma = 8\%$
Portfolio B generates a return of 12%, $\sigma = 5\%$

If $R_f = 5\%$: Sharpe ratio A = 1.25
Sharpe ratio B = 1.4

Portfolio Expected Returns

The expected return of a portfolio is the weighted average of the expected returns for each asset in the portfolio:

$$E(R_p) = \sum_{i=1}^n w_i E(R_i)$$

w_i = % of portfolio invested in each asset

Capital Market Line (CML)

By adding a risk-free asset, we generate a linear risk/return trade-off

This line, the CML, provides a better risk/return trade-off than the efficient frontier

Optimal Investment Combinations

Consider a 50/50 investment in a portfolio of risky assets and the risk free asset that maximizes Sharpe Ratio

What is the expected return and standard deviation of this new portfolio?

(should be lower return/lower risk)

Standard Deviation

The variance of the risk-free asset, and the covariance of the risk-free asset with any other asset, are zero!

Std. deviation of a portfolio (comprised of a risky portfolio "a" & and risk-free asset) is simply:

$$\sigma_p = w_a \sigma_a$$

Return Variability

Variance:

=var.s()

$$\sigma^2 = \frac{\sum_{i=1}^n (R_i - \bar{R})^2}{n-1}$$

(where "n" = number of returns)

Standard Deviation:

=stdev.s()

$$\sigma = \sqrt{\sigma^2}$$

Covariance

$$\text{Cov}(X, Y) = \frac{\sum_{i=1}^n (X_i - \bar{X})(Y_i - \bar{Y})}{n-1}$$

=covariance.s(array1, array2)

Measure the deviation of returns from their average and multiply the deviations